

Ian Leung

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WORK EXPERIENCE

Roblox

Jun 2026 – Aug 2026

Software Engineer Intern (Network Reliability) | Go, Python

San Mateo, CA

- Automated the Network Engineer deployment process, pushing configs to **5,000+** devices in global datacenters
- Built a GitHub Actions CI/CD pipeline to eliminate manual tasks, saving **30+ engineering hours** per month
- Orchestrated Temporal Workflows & LLM analysis for safe, autonomous deployment of network changes, preventing outages/misconfigurations and identifying issues on **20+** network devices
- Identified and fixed a resource leak in an internal network service, recovering **7%** CPU utilization across devices

WQC

Jan 2026 – Present

Software Engineer (DevOps) | Rust, Python, AWS

Waterloo, ON

- Engineered portal.waterlooquantclub.com, a full-stack portal hosted on AWS for **700+** members, supporting JWT authentication, custom application forms, resume uploads, reviewer workflows, and Excel exports
- Solo-developed a Grafana observability platform using PostgreSQL, TLS, and AWS, providing real-time monitoring of **100+** metrics and automatic **Discord bot alerting** for critical service failures
- Built and deployed [2v2bluff](#), a real-time multiplayer **Rust/WebSocket** card game supporting four concurrent players

Sunnybrook Research Institute

Jul 2025 – Aug 2025

Software Research Intern | Python

Toronto, ON

- Built an event tracing and playback tool to simulate medical professionals' interactions with **Focused Ultrasound** software, enabling realistic testing and diagnosis of software defects, unintended user behaviors, and edge cases
- Created an automated software testing pipeline that generates synthetic and replayable JSON event flows, reducing software testing times by **91%**
- Developed a logging service to validate test results against baselines, catching **4** software defects before production release

PROJECTS

Pharmagotchi – Pharmacy Companion App | Android Studio, Figma, Kotlin, XML, OpenRouter

- Built a Tamagotchi-inspired mobile health-companion app to track medications and health, backed by an offline Room database and WorkManager background reminders
- Integrated an **OpenRouter**-powered chat assistant with Retrieval-Augmented Generation (RAG) that parses health conditions and medications into structured JSON vitals and provides context-aware medication guidance
- Connected Fitbit tracking devices through the Google Health API, centralizing health data for seamless cross-device monitoring

PeerAssist – Peer Review Platform | Java, Supabase, Apache PDFBox, Maven

- Created a full-stack **Java** desktop application for **20+** students to upload, review, and grade peers' academic work
- Integrated a **Supabase** backend for authentication and data storage, enforcing **Row-Level Security** policies across profiles, documents, and reviews tables to control per-user access
- Rendered uploaded PDFs with Apache PDFBox and added search, sorting, and ranking by average review score; packaged the app into a single zero-config executable JAR via **Maven** shade plugin

SKILLS

Languages Python, Java, C++, C#, C, Go, Kotlin, TypeScript, SQL, HTML, CSS, Bash

Technologies Temporal, gRPC, REST, PyTorch, Pandas, Grafana, PostgreSQL, MongoDB, React, Actions

Tools Linux, Git, Docker, AWS (EC2, S3, Lambda, etc.), OpenRouter, Jupyter, Supabase, Android Studio

EDUCATION

University of Waterloo

GPA: 3.8/4.0

Bachelor of Computer Science, Finance, & Statistics – Triple Major

- President's Scholarship of Distinction (**\$5000**), Undergraduate Achievement Award (**\$1500**)
- Relevant Coursework: Object-Oriented Programming (Advanced), Sequential Programming (Advanced), Functional Programming (Advanced), Interpreters (Advanced), Linear Algebra, Data Analytics